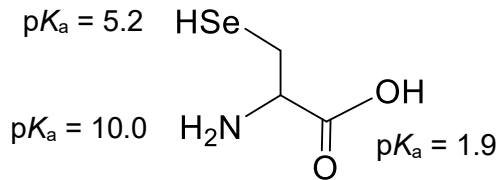


- 15** The structure of an amino acid, Selenocysteine, with its three functional groups is shown below. The pK_a of $-\text{SeH}$, $-\text{NH}_2$ and $-\text{COOH}$ are 5.2, 10.0 and 1.9 respectively.



Which statements are correct about Selenocysteine at fixed temperature?

- 1 The carboxylic acid group ($-\text{COOH}$) is the most acidic group.
- 2 At pH 8, Selenocysteine will migrate towards the positive terminal in an electric field.
- 3 When the concentration of H^+ is high, K_a of the three functional groups decreases due to lower extent of dissociation.

A 1 only **B** 1 and 2 only **C** 1 and 3 only **D** 1, 2 and 3